

GISMA UNIVERSITY OF APPLIED SCIENCES

B141 Data Mining

Individual Project Report

Topic :- Starbucks Customer Behavior and Promotional Offer Analysis Using the Data Mining techniques

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Github Repository Link :-

Notice : Due to GitHub's limitation to upload file more than 25 MB, the files master_data.csv, transcript.csv, and transcript_cleaned.csv are not included in the GitHub repository. These files can be regenerated by executing the Jupyter notebooks provided in the repository, and the original dataset is available from the referenced Kaggle source.

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1. Introduction

In this modern era, the fast growth of digital technologies has eventually resulted in the institutions that is collecting huge amount of data by business operations on daily basis, transactions made by customers, online platforms and then marketing campaigns. Irrespective of the fact that this data actually has a lot of information, yet this data can't directly provide the useful insights without analysing it properly. And here comes the Data Mining, which is the process through which one can discover useful patterns, relationships and knowledge obtained from huge datasets using computational and statistical techniques (Han, Pei and Tong, 2022). Through this, companies can convert the raw data into actionable insights that support decision making and improve the performance of business.

In the retail industry, because of growing competition and changing preferences of consumers, understanding customer behaviour is highly important. Data Mining techniques helps the companies to understand the hidden trends which one can not see with normal data, evaluate promotional campaign performance, distributing consumers into meaningful groups, and create a system that recommends and delivers personalised experiences.

Starbucks is one of the largest global coffee chain and it also operates the customer membership programme which provides different types of promotional campaigns that includes discounted offers, Buy one get one (BOGO) promotions and international offers. When these sort of interactions are analysed, it helps in identifying what sort of offers performs well, how different customer groups respond to marketing campaigns and how the promotions for future can be improved.

In this project, the concepts and techniques are applied to Starbucks customer dataset which were learned in the module B141 Data Mining. A pipeline is followed in this project which starts with data understanding moving further to data preprocessing, data integration, analysing the data, segmenting the customers into groups and then performing the recommendations. Customer Segmentation is done with the help of factors such as income and age.

2. Project Objective

The main objective of this project is the application of techniques of data mining to analyse the behaviour of customers and promotional campaign data from the Starbucks dataset.

Another goal of this project is to perform analysis so that we can identify trends and relationships within the data. Customers are distributed according to demographic characteristics like age and income for the better understanding of different groups and their potential marketing needs.

Eventually, the recommendation system has been made at the end with the help of collaborative filtering and cosine similarity. By analysing the similarities between promotional offers based on customer interaction, recommendations are generated by the system that gives more relevant promotions to customers.

3. Dataset Description

The Dataset used in this project is actually obtained from Kaggle (Muliar, 2023). Promotional offer dataset of Starbucks is used for this project which is actually publicly available and contains information regarding the demographics of customers, promotional offers, and interactions made by customers. There are three separate files in this dataset named 'profile.csv', 'portfolio.csv' and 'transcript.csv'. Each single dataset gives unique information which when integrated provides a broad overview of customer behaviour and promotional campaign performance.

3.1 Profile Dataset

This dataset has demographic data for Starbucks customers. It has customer identifies, gender, age, annual income and the date of membership.

The values like age, gender and income are also used for segmenting the customers and behavioural analysis.

This dataset has 17000 rows and 5 columns, making it main source of customer demographic data.

3.2 Portfolio Dataset

The portfolio dataset has the information about promotional campaigns that are held by Starbucks. Each promotional offer has uniqueness with the factors such as reward value, level of difficulty, time duration, channels to communicate and offer type.

The offer identifier is then extracted from the third dataset named Transcript dataset which is used to merge promotional information into the main integrated dataset.

There are 10 rows and 6 columns in this file through which 10 different promotional offers are represented.

3.3 Transcript Dataset

This file has the information regarding customer activities throughout promotional campaigns. Every record tells an event which is generated by a customer, including offer received, offer viewed, offer completed and normal purchase transactions.

Unlike other two datasets, additional information is stored by transcript dataset in nester value column. This is the reason why preprocessing is important to extract important fields like identifiers and transaction amounts before integration.

There are 306,534 rows and 4 columns in this dataset, making it the largest used dataset in this project.

4. Methodology

A well – organized methodology is followed in this project to transform raw Starbucks customer Data into meaning business insights. It consists of various stages starting with understanding the Data and ending with the development of recommendation system. Each step is dependent upon the previous step so that it ensures data is properly prepared, analysed and interpreted before recommendations are generated.

The project actually starts with data understanding, where the structure, dimensions, attributes and dataset's quality is examined.

Furthermore, preprocessing is done in second stage. In this stage, datasets are cleaned by handling missing values, and then correcting their inconsistencies, replacing the values that are invalid and finally preparing data for further analysis.

After previous step, datasets are integrated into one master file using common columns.

Master dataset is then used for analytics for identifying customer behaviour patterns, performance of promotional campaigns, and relationships between different sort of variables.

Customer Segmentation is performed to divide customers into meaningful groups based on demographic characteristics like age and income.

At last, we have recommendation system which is developed using collaborative filtering and cosine similarity. Customer interactions is used to create customer off matrix after which cosine similarity is used to calculate to measure the similarity between promotional offers. Based on these similarity scores, the system recommends promotional offers which are closely related according to the behaviour of the customers.

4.1 Data Understanding

Initial step of the project is to understand the structure and quality of the Starbucks datasets before preprocessing or analysis are performed. The datasets named as 'profile.csv', 'portfolio.csv' and 'transcript.csv' were explored in starting to understand their contents, dimensions, data types and also the relationships. This step is actually very important because understanding the data thoroughly enables identifying the potential issues and ensures that preferred preprocessing techniques are selected.

These three datasets were loaded in python using Pandas library and it was initially inspected using functions like head(), shape, columns, info() and describe(). Several information like number of rows and columns, attribute names, data types, and summary statistics were provided by using these functions.

Missing values and inconsistencies were identified using function such as isnull().sum() by which it was known if any attribute contained missing information that would require preprocessing. And here it was noticed that customer profile dataset contained some missing values in variables like gender and income. However on other hand ages that were recorded were false.

Transcript file was also examined carefully because it stored additional information inside a nested volume instead of separate attributes. It was necessary to understand this structure because these nested information was supposed to be extracted later before the datasets could be merged.

In total, this stage provided a clear overview of available information. Also it provided the data quality issues and made a relationship between three datasets. This formed the base of further processes.

4.2 Data Preprocessing

After understanding the data, it is important to preprocess the data. In this stage data quality was improved by handling missing values, correcting inconsistencies and then making the datasets for integrating and analysis. Preprocessing should be handled with care because incomplete or inaccurate data can affect negatively the quality of final results.

It begins with profile dataset. Pandas functions were used to identify the missing values which helped in detecting incomplete information. There were missing values in gender attribute which were then replaced by "Unknown" value. Similarly for missing income values, they were replaced by their median value.

In this process, ages of the customers were also examined. It was seen that for many customers the age written was 118, which showed missing or unavailable age information instead of actual age of the customer. Before proceeding further, these values were replaced with missing values (Nan).

It actually required small preprocessing because it was already organized well. The communication channels which were linked with each promotional offer were stored as lists, making them less suitable for analysis. These were converted into readable texts using the functions of python.

The transcript dataset needed more preprocessing because necessary data like promotional offer identifiers and transaction amounts was stored inside nested value column. The `ast.literal_eval()` function was used to convert the text into dictionaries through which these values could be extracted as different column. Eventually, factors like `offer_id` and the amount could be accessed directly for analysis and integration.

At the end cleaner version of the files were saved as CSV files.

4.3 Data Integration

After cleaning and preparing the data, the data is to be integrated which will be further used for analysing. Because of the fact that all the necessary data like customer information, promotional offer details and records of customer interaction were recorded in different files, it was important to get an overview of customer behaviour.

This process was actually started with transcript file. The promotional offer identifier is used as common key to combine transcript and portfolio dataset.

The resulting dataset was then combined with customer dataset with the help of person identifier which represents the customers uniquely.

Now, all the important attributes like customer, promotion and transaction were in same table, which made it easier to perform the behaviour analysis, segmentation and recommendation.

The master data was exported as `master_data.csv`

4.4 Customer Behaviour Analysis

After integrating the data successfully, data was analysed to to identify patterns, relationships and trends in the Starbucks customer Data. This part of the project focused on understanding behaviour of customers, promotional offer performance and demographic characteristics with the help of statistics and data analysis techniques.

So here is how it begins, distribution of customer events which were recorded in the dataset was examined. All types of event types were analysed how customers reacted to promotional offers.

The distribution of promotional offer types was also analysed to see how different marketing strategies were used by Starbucks. The Dataset had different categories of promotional offers which included discount offers, buy one get one (BOGO) and informational campaigns.

Demographics characteristics of customers were analysed by analysing the attributes like age, gender and income. This provided better view of customer population and reflected the overall demographic composition of dataset.

For further investigations between numerical variables, a correlation analysis was performed between age and income of the customer. This helped in understanding if relationship existed between these variables and at the same time, it also provided additional insight into customer demographics.

4.5 Customer Segmentation

Distributing the customers was performed in this step as they were divided into some meaningful groups based upon their age, income. This steps actually helps business to understand different categories of the customers and supports the development of more targeted marketing strategies.

Customers were mainly segmented with the help of two variables age and income. Age based segmented customers into three categories Young, Middle Age and Senior. In the same way, income based segmentation classified customers into three categories which are Low income, Medium income and High income. These categories were actually created using simple rule – based conditions which assigned each customer to their respective segment based on predefined thresholds.

After assigning customers to the meaningful groups, the distribution of each segment was analysed to understand the composition of customer population.

4.6 Recommendation System

This is the final stage of this project which focused on developing a recommendation system which can actually identify similar promotional offers based on customer behaviour using collaborative filtering (Aggarwal, 2015). And in this project, a collaborative filtering approach was selected as it items are recommended by it by analysing the patterns in customer interactions instead of just trusting the characteristics of the promotional offers.

This process starts by creating a Customer-offer matrix. This matrix has rows and columns which represented customer and promotional offer respectively and the values tells if the customer had completed a particular offer or not. This matrix converted customer interaction data into a structured format suitable for similarity analysis.

Cosine similarity was applied to the matrix after it was created to measure the similarity. Cosine similarity calculates how closely two offers are related based on the interaction of the customer with promotion. Offers with high similarity scores indicate that they were frequently completed by same sort of group however on the other hand the one with lower score had weaker relationships.

The similarity values were organised into an offer similarity matrix which had similarity score between every single pair of the offers.

At the end, this system identified top similar offers for a selected promotional offer by sorting the similarity scores in descending order. The offer always got similarity score of 1 to itself while others were ranked according to the similarity values. These recommendations tells how collaborative filtering can help organisations in identifying related promotional campaigns which may fit for customers with same behavioural patterns.

5. Results and Discussion

The analysis reflected that customers did the transactions the most which indicates that customers interacted with Starbucks mostly through purchases than promotional campaigns.

Customer demographic Analysis gave a better image regarding characteristics of customer base. The preprocessing step made sure that missing values and invalid records were handled with caution which resulted in cleaner and better data. The correlation analysis between age and income gave further insights into the relationship between these variables and supported the segmentation process.

Furthermore, customer segmentation process divided customers into meaningful groups based on their income and age. These segments allowed institutions to understand customer groups and give opportunities for more targeted promotional campaigns.

This system resulted in one of the most important outcomes of the project. By using collaborative filtering with cosine similarity, relationship between promotional offers were successfully identified. The matrix which was generated at the end demonstrated that some offers share more stronger relationships than others which indicated that customers those who have completed one offer completed similar offers. The top similar offers created by the recommendation system states how institutions can suggest related promotions with respect to previous customer interactions.

6. Conclusion

Data mining techniques were successfully applied in this project to analyse customer behaviour and promotional data with the help of using Starbucks data. Started with data understanding and preprocessing, the data were cleaned , integrated and was made better for analysing the data.

Customer behaviour analysis gave the valuable insights into promotional campaign performance and customer demographics. And on the other hand, customers were grouped based on their age and income in customer segmentation. These segmentation explained how this information can actually support more targeted marketing strategies and improve customer understanding.

At the end of the project in the last notebook, the recommendation project was created using cosine similarity and collaborative filtering. By generating the customer-offer matrix and calculating similarity scores between promotional offers, the system successfully identified related offers which suggested to the consumers based on their previous interactions with the company. This actually showed the real use of recommendation system within a real business scenario.

In total, this project shows that how can data mining techniques transform raw data into meaningful information that supports decision making for the company. Several steps like preprocessing, integration, analysis, segmentation and recommendation provided a organised approach with which one can understand customer behaviour and improve promotional strategies.

7. References

Aggarwal, C.C. (2015) *Data Mining: The Textbook*. Cham: Springer.

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